

Candidate Model Shortlist

Deep learning (CNN), transformer, and hybrid architectures to explore on the merged six-class ASD corpus. Shaded rows are models already in your benchmark.

All timm identifiers verified against timm 1.0.29 · 30 August 2026

1. Deep learning — convolutional

Already in your benchmark

MODEL	TIMM IDENTIFIER
VGG-16	vgg16
ResNet-50	resnet50
DenseNet-121	densenet121
Inception V3	inception_v3
EfficientNet-B0	efficientnet_b0
MobileNetV2	mobilenetv2_100

Recommended additions

MODEL	TIMM IDENTIFIER
ConvNeXt-V2 Base	convnextv2_base.fcmae_ft_in22k_in1k
ConvNeXt-Base (DINOv3)	convnext_base.dinov3_lvd1689m
DenseNet-201	densenet201
Inception-ResNet-V2	inception_resnet_v2
SE-ResNeXt-50	seresnext50_32x4d
RegNetY-3.2GF	regnety_032

2. Transformer

Already in your benchmark

MODEL	TIMM IDENTIFIER
ViT-Base	vit_base_patch16_224
Swin-Base	swin_base_patch4_window7_224
Swin-Tiny	swin_tiny_patch4_window7_224
DeiT-Small	deit_small_patch16_224

Recommended additions

MODEL	TIMM IDENTIFIER
DINOv3 ViT-B/16	vit_base_patch16_dinov3.lvd1689m

DINOv3 ViT-L/16	vit_large_patch16_dinov3.lvd1689m
DINOv2 ViT-B/14	vit_base_patch14_dinov2.lvd142m
EVA-02 Base	eva02_base_patch14_224.mim_in22k
DeiT-III Base	deit3_base_patch16_224.fb_in22k_ft_in1k
Swin-V2 Base	swinv2_base_window12to16_192to256.ms_in22k_ft_in1k
BEiT-3 Base	beit3_base_patch16_224.in22k_ft_in1k
Hiera Base	hiera_base_224.mae_in1k_ft_in1k
SigLIP ViT-B/16	vit_base_patch16_siglip_224.webli

SigLIP is included as a negative control, not as a candidate. Language-supervised encoders localise faces but discriminate facial parts poorly — roughly 25 points behind DINOv3 on facial features specifically.

3. Hybrid CNN–Transformer

These are the control condition for a proposed hybrid architecture, not optional extras. Neither VGG-16 nor Swin is a hybrid, so nothing currently in the benchmark answers the question "why not use an existing hybrid?"

MODEL	TIMM IDENTIFIER
CoAtNet-2	coatnet_rmlp_2_rw_224.sw_in12k_ft_in1k
MaxViT-Tiny	maxvit_tiny_tf_224.in1k
MaxViT-Small	maxvit_small_tf_224.in1k
CAFormer-S36	caformer_s36.sail_in22k_ft_in1k
ConvFormer-S36	convformer_s36.sail_in22k_ft_in1k
MobileViT-V2	mobilevitv2_100.cvnets_in1k
EfficientFormer-V2-S2	efficientformerv2_s2.snap_dist_in1k
FastViT-MA36	fastvit_ma36.apple_dist_in1k

4. Expression-specialised models

External checkpoints rather than timm; budget roughly one day of integration each. Worth it because the weights already encode expression rather than generic objects.

MODEL	SOURCE
POSTER++	github.com/talented-q/poster_v2
DAN	github.com/yaoing/DAN
APViT	github.com/youqingxiaozhua/APViT
EmoNet	github.com/face-analysis/emonet
ResMaskNet	github.com/phamquilluan/ResidualMaskingNetwork
SwinFace	github.com/lxq1000/SwinFace
LA-Net	github.com/crazy-hbc/LA-Net

5. Face-identity and face-SSL encoders

MODEL	SOURCE
ArcFace-R100	InsightFace model zoo
FaRL	github.com/FacePerceiver/FaRL
FSFM	arXiv:2412.12032
MARLIN	github.com/ControlNet/MARLIN

ArcFace has no expression supervision at all, so it doubles as a leakage probe: if it predicts emotion well on your splits, the likeliest explanation is that identity is still leaking between train and test.

Priority order

Ranked by expected payoff per GPU-hour at your data scale (~1,700 development images).

#	EXPERIMENT	NOTE
1	Expression-domain pretraining	ImageNet → AffectNet → ASD corpus. Larger effect than any architecture swap; request AffectNet access first, approval is slow.
2	Frozen DINOv3, intermediate depth	Probe blocks at ~60/75/100% of depth, not the final block alone. Correspondence degrades in the last block.
3	Training regime comparison	Full fine-tune vs LoRA vs multi-depth probe vs linear probe, on the top two backbones.
4	Hybrid controls	CoAtNet, MaxViT, CAFormer. Required for the hybrid contribution to survive review.
5	Action-unit + asymmetry baseline	OpenFace or Py-Feat features into gradient boosting. A go/no-go on the hemiface hypothesis before building it.
6	FER specialists	DAN (AffectNet checkpoint) and POSTER++ first.
7	ArcFace leakage probe	Under an hour; evidence that the splits are clean.

Not worth exploring

- Vision-language models (SmolVLM, PaliGemma, BLIP-2, FaceLLM) — all skew heavily toward happiness with near-zero F1 on negative emotions, which is precisely where your rare classes sit.
- Larger variants of models already tested (Swin-Large, ViT-Large, ResNet-152) — more capacity against 1,700 images moves the overfitting curve, not the result.
- A newer generic backbone — the ten already tested sit inside one another's confidence intervals. This is a data-scale limit, not an architecture gap.
- Heavy augmentation (MixUp, CutMix) — blending faces destroys the subtle asymmetries that motivate the study.

One augmentation to reconsider: horizontal flip is still in the pipeline. It imposes a left-right symmetry prior on a dataset whose premise is that autistic facial expression is less symmetrical. Ablate with and without it, and report the result.